SCATS LEARNING LAB · STUDY GUIDE

Five new ideas for SCATS, tested honestly

Connected cars, a learning agent, max-pressure, broken detectors and signal prediction: what each one does, how to test it fairly, and what really happened in SUMO.

Level 3 · Mastery

INSIDE

- How to judge a new idea fairly: change one thing, use several seeds
- Why 1 connected car in 20 is not enough to measure traffic
- A learning agent that lost to SCATS, and why that is useful
- How three in ten broken loops raised delay by 60%, and how to recover

Beginner-friendly companion to smrahmanbd.github.io/scats-learning-lab

The map: SCATS Mastery

This is the research level, written for beginners. You should know what SCATS is (Foundation) and how its cycle, split and offset logic works (Advanced), but every idea is explained again in plain words. There are six ideas: one on how to test a new idea, then five research directions, each implemented in this lab's SUMO controller. Several did not beat standard SCATS. That is normal in research, and this guide shows you why.

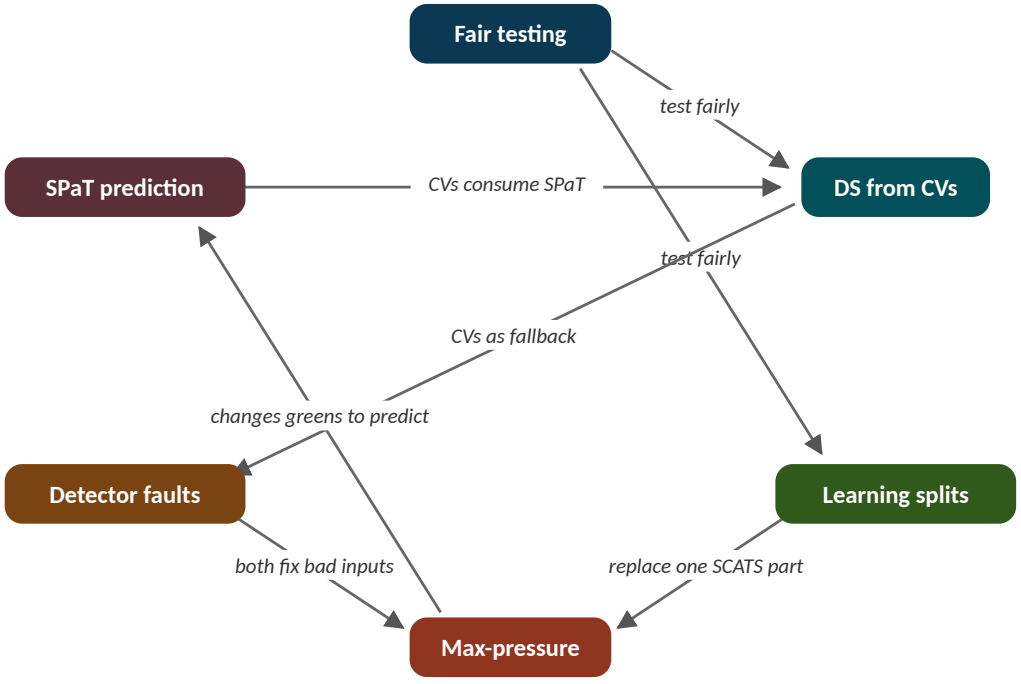


Figure 0. The whole guide on one page. Each box is one concept section; the arrows say how one idea feeds the next. Come back to this page whenever you feel lost.

1. How to evaluate a new SCATS idea
2. Degree of saturation from connected vehicles
3. Learning split selection inside the SCATS envelope
4. SCATS and max-pressure
5. Resilience to detector faults
6. Predicting signal timing for connected vehicles

HOW TO USE THIS GUIDE

Read one concept per sitting (20 to 30 minutes). Do every **Simple calculation** with a pencil before reading the result. Say the **Picture this** scene out loud: the sillier, the better. The **Recap so far** box grows as you go and is your free revision. When you finish, take the self-quiz, then follow the revision plan on the last page. If you want to run the experiments, every concept names its script in the `sumo/mastery/` folder.

What you will be able to do

- Design a fair test of a new signal-control idea: one change, three baselines, several seeds, more than one measure.
- Estimate degree of saturation (DS) from connected-vehicle counts and explain why low penetration gives noisy estimates.
- Do one Q-learning update by hand and explain why the lab's agent picks only engineer-approved split plans.
- Compute max-pressure greens and blend them with SCATS greens using a weight β.
- Explain what stuck-on and stuck-off loops do to DS, and how alarms and imputation recover performance.
- Compare signal-timing predictors by mean absolute error and say why an adaptive controller is hard to predict.

Before we start

HOOK

Two of the five new ideas lost. A reinforcement-learning agent and a max-pressure hybrid both gave more delay than standard SCATS on this lab's corridor. Reporting that honestly is what makes the other results believable.

HOOK

Three broken loops in ten raised delay by 60%. When 30% of the detectors failed, mean delay jumped from 36.9 to 59.2 seconds per vehicle. Two simple alarm rules brought it back to 40.7 s.

HOOK

A connected car can mislead you by being lucky. If only 1 car in 20 reports its position, one green may contain zero reporting cars and the next may contain three. With 5% penetration the DS estimate was off by 0.65 on average: bigger than many real DS values.

Words you will meet

You do not need to memorise these now. Each one is explained again when it first matters.

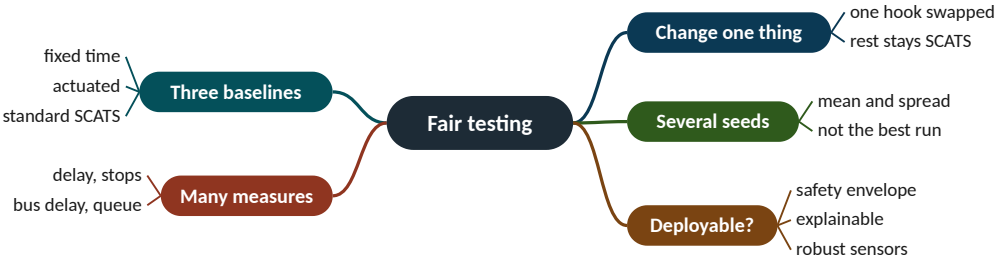
Word	In plain words
Degree of saturation (DS)	How fully a green was used: 1.0 means the green was busy from start to end. SCATS measures it at stop-line loops.
Split plan	A stored way of sharing the cycle between phases, for example 'main road 55%, side road 45%'.
Baseline	The method you compare against, for example fixed time, actuated, or standard SCATS.
Seed	A number that fixes the random traffic in a simulation. Different seeds give different, equally likely days.
Connected vehicle (CV)	A car that reports its position and speed by radio or mobile data.
Penetration rate p	The share of vehicles that are connected: p = 0.25 means one in four.
Reinforcement learning (RL)	A program that learns by trial and error, getting a reward after each choice.
Max-pressure	A control rule that gives green to the movement with the most waiting traffic and the most room to go to.
SPaT	Signal Phase and Timing: a message telling cars which light is on and when it will change.

CONCEPT 1 OF 6 · SCATS MASTERY

How to evaluate a new SCATS idea

HOOK

A new controller beats SCATS by 3 seconds in one simulation run. Is it better? You cannot tell yet. Here is why, and what to do instead.



IN PLAIN WORDS

Testing a new running shoe? Change only the shoe, not the shoe and the track. Race on several days, not one lucky windy day. And time the sprint and the marathon, because a shoe can help one and hurt the other. *Where the picture breaks:* a runner can feel tired; a simulation seed is exactly repeatable, which is what lets us compare two controllers on the same traffic.

Research on traffic control is full of big claimed gains that disappear in practice. Most of them break one of a few simple rules. The first rule is to **change one thing**. In this lab every new idea plugs into SCATS through one **hook** (a slot in the controller settings): `ds_source`, `split_selector`, `green_allocator` or `sensor_filter`. Everything else stays standard SCATS, so any change in delay is caused by the new idea.

The second rule is to compare against **three baselines** on identical demand: fixed time, actuated and standard SCATS. The third is to use **several seeds**. A seed fixes the random arrivals; a new seed is a new, equally likely day. Report the mean and the spread across seeds, never your best run.

The fourth rule is to report **more than delay**: stops, bus delay and queue tell different stories, and a method can win one by losing another. Finally, ask what would stop it being used on a real street: safety limits, whether an engineer can explain a decision, what data it needs, and how it behaves when sensors fail.

$$\text{difference per seed } d_s = \text{delay}_{\text{new}} - \text{delay}_{\text{SCATS}}$$

same seed, same traffic, so the difference is fair

$$\text{report mean}(d) \text{ and spread (standard deviation) of } d$$

a mean much bigger than the spread is a real effect

SIMPLE CALCULATION · Is the RL agent worse than SCATS?

Mean delay (s/veh) on three unseen seeds. SCATS: 37.91, 35.57, 35.86. RL agent: 39.37, 42.27, 40.27.

1. Differences: 39.37 − 37.91 = **1.46**, 42.27 − 35.57 = **6.70**, 40.27 − 35.86 = **4.41**.
2. Mean difference: (1.46 + 6.70 + 4.41) ÷ 3 = 12.57 ÷ 3 = **4.19 s**.
3. Spread (sample standard deviation): about **2.63 s**.
4. All three differences are positive, and the mean is larger than the spread.

The agent is worse by about 4.2 s per vehicle on every seed. One seed alone could have shown anything from 1.5 to 6.7 s.

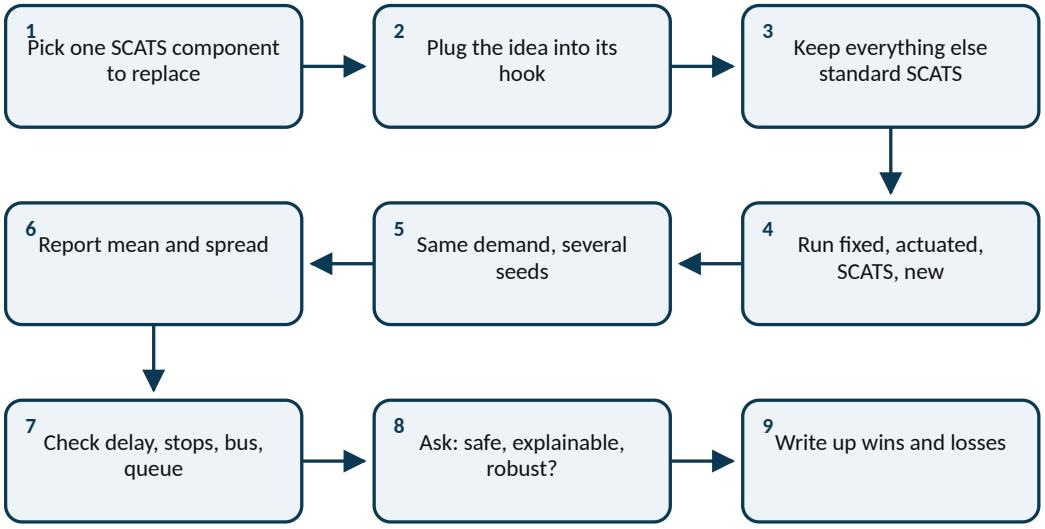


Figure 1. A fair test of a new idea. What to notice: the new idea enters through one hook, it is compared with three baselines on the same seeds, and it must pass a deployment check, not only a delay check.

TRAP

Do not tune your new method on the same seeds you use to test it. The RL agent here was trained on other seeds and tested on unseen seeds 7, 8 and 9. Testing on training seeds flatters the method.

PICTURE THIS · STOP 1 OF THE ROUTE

At the lighthouse the keeper races four identical toy ships in a bathtub. She changes only one sail on the new ship, then throws three dice to make three different storms, and writes every result on the wall, even the races the new ship lost.

What it encodes: One sail = change one thing; four ships = three baselines plus the new idea; three dice = several seeds; writing every result = report mean, spread and losses.

TRY IT NOW (about 90 seconds)

A new method is tested on two seeds. Its delay differences from SCATS are −2.0 s and +1.0 s. What is the mean difference, and can you claim it is better? *Answer in the key at the back.*

RECAP SO FAR

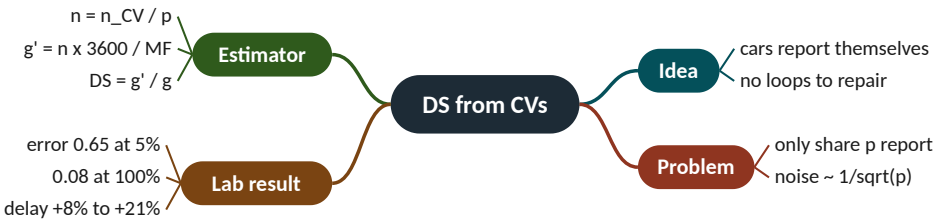
1. **Fair testing: change one hook, compare with fixed, actuated and SCATS on the same seeds, report mean and spread of several measures, check deployability.**

CONCEPT 2 OF 6 · SCATS MASTERY

Degree of saturation from connected vehicles

HOOK

Loops cut into the road are expensive and break often. Cars already carry phones and GPS. Could SCATS measure traffic from the cars instead?



IN PLAIN WORDS

It is like guessing how many people came to a party from the guests who posted photos. If one guest in four posts, you multiply by four. With only a few posters, one extra photo changes your guess a lot. *Where the picture breaks:* posters are not random: some people always post. Connected cars may also not be a random sample (for example, newer cars), which this simple estimator assumes away.

SCATS normally measures **degree of saturation (DS)** at stop-line loops: how much of the green was used by moving vehicles. A **connected vehicle (CV)** reports its own position, so the controller can count CVs that cross the stop line during a green. Only a share **p** (the **penetration rate**) of vehicles is connected.

The simplest estimator scales the count up by 1/p, then converts vehicles into the green time a saturated stream would need, using the lane's maximum flow MF (vehicles per hour of green). Dividing by the actual green gives an estimated DS.

The catch is sampling noise. Whether a car is connected is like a coin toss, so the count of CVs jumps around, and dividing by a small p magnifies the jump. In shadow mode (same traffic, estimate compared with loop DS over 2,122 lane-cycles) the mean error was 0.65 at p = 5% and 0.08 at p = 100%. Driving SCATS with the estimate gave 8% to 21% more delay than loops.

$$\hat{n} = n_{CV} \div p \quad g' = \hat{n} \times 3600 \div MF \quad \hat{DS} = g' \div g$$

count, scale up, convert to green needed, divide by green given

$$\text{error} \propto 1 \div \sqrt{p}$$

halving the error needs four times the penetration

SIMPLE CALCULATION · One green at 25% penetration

Green g = 30 s. MF = 1,800 veh/h, so a saturated stream needs 3600 ÷ 1800 = 2 s per vehicle. In truth 14 vehicles crossed. 4 of them were connected. p = 0.25.

1. Scale up: $\hat{n}$ = 4 ÷ 0.25 = **16 vehicles** (truth: 14).
2. Green needed: $\hat{g}$ = 16 × 2 = **32 s**. Estimated DS = 32 ÷ 30 = **1.07**.
3. If counted perfectly: 14 × 2 = 28 s, DS = 28 ÷ 30 = **0.93**.
4. Spread of the CV count: √(14 × 0.25 × 0.75) = **1.62** cars, so $\hat{n}$ wobbles by 1.62 ÷ 0.25 = 6.5 cars, and DS by 6.5 × 2 ÷ 30 = **0.43**.
5. Chance that none of the 14 is connected: 0.75^{14} = **0.018**, about 1 green in 56.

One lucky extra CV moved DS from 0.93 to 1.07. Typical wobble is ±0.43, which is why SCATS on 25% CV data cannot match loops green by green.

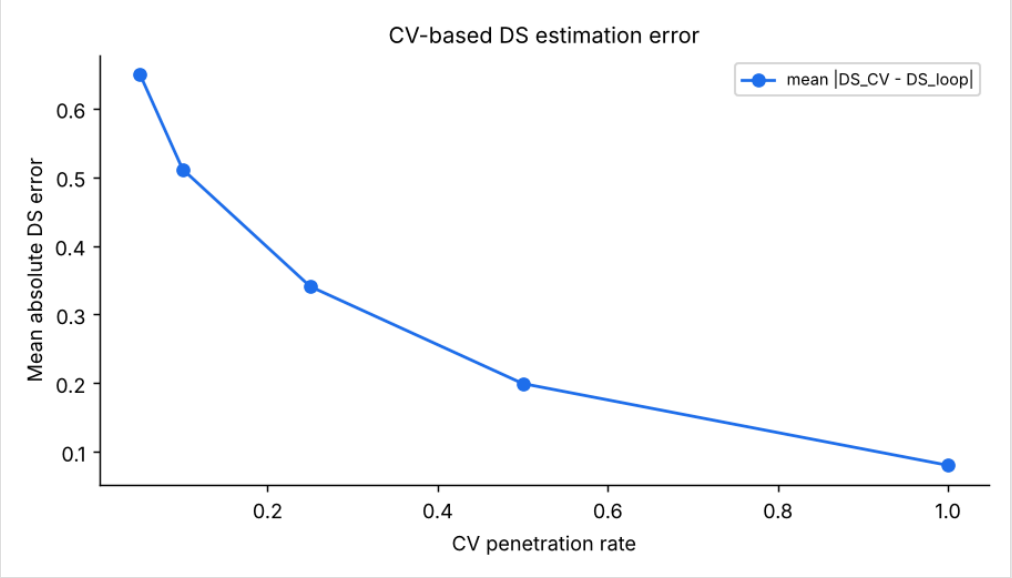


Figure 2. Mean DS error against CV penetration (shadow mode, 2,122 lane-cycles). What to notice: error falls fast at first, then slowly, roughly like 1/√p: about 0.65 at 5%, 0.34 at 25%, 0.20 at 50%, 0.08 at 100%. Even at 100% it is not zero, because counting cars cannot see gaps in how the green was used.

TRAP

Do not expect delay to improve smoothly as penetration rises. In closed loop, delay was +11% at 5%, +8% at 25% and +21% at 50%. SCATS's plan voting absorbs some noise, and noisy overestimates lengthen greens, which can help buses and stops by chance.

PICTURE THIS · STOP 2 OF THE ROUTE

On the quay, the harbourmaster counts returning fishing boats by radio. Only one boat in four has a radio, so every 'beep' he hears he multiplies by four on a huge chalkboard. A seagull steps on the radio, one extra beep sounds, and his total leaps by four boats.

What it encodes: Radio boats = connected vehicles; one in four = p = 0.25; multiply by four = $\hat{n} = n_{CV}/p$; the seagull's extra beep = sampling noise magnified by 1/p.

TRY IT NOW (about 90 seconds)

p = 0.5, green 30 s, MF 1,800 veh/h, and 9 CVs crossed. What is the estimated DS? *Answer in the key at the back.*

RECAP SO FAR

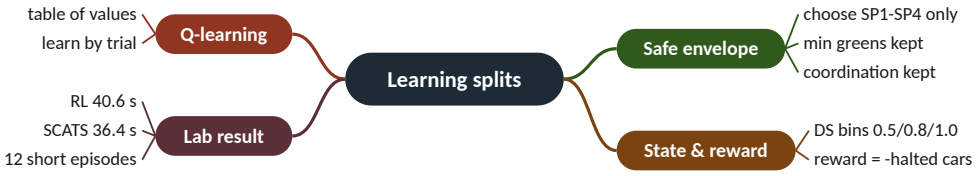
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2. **DS from CVs: $\hat{n} = n_{CV}/p$, $g' = \hat{n} \cdot 3600/MF$, DS = g'/g; noise $\propto 1/\sqrt{p}$; error 0.65 at 5%, 0.08 at 100%; closed-loop delay +8% to +21%.**

CONCEPT 3 OF 6 · SCATS MASTERY

Learning split selection inside the SCATS envelope

HOOK

Papers often report that a learning agent cuts delay by 20% or more. Few such agents run on real streets. What if the agent could only choose among plans an engineer already approved?



IN PLAIN WORDS

Think of a new waiter allowed to recommend only dishes from the chef's menu. He learns from tips which dish suits which table. He cannot invent a dangerous dish, but early on he still recommends worse than the experienced head waiter. *Where the picture breaks:* a waiter gets a tip at once; the agent's reward (fewer stopped cars) is noisy and arrives after a whole cycle, so learning is slow.

Reinforcement learning (RL) learns by trial and error. In each situation (the **state**) it picks an **action**, receives a **reward**, and updates its estimate of how good that action is. Deep RL controllers can, in principle, do anything with the lights, which makes them hard to certify and explain.

This lab restricts the agent to the SCATS **envelope**: it only replaces the split selector, choosing among four engineer-approved split plans SP1 to SP4. Minimum greens, intergreens, cycle length and offsets stay standard SCATS, so the agent cannot make an unsafe timing.

The agent is **tabular Q-learning**: a table of values Q(state, action), shared by the four intersections. The state is each phase's DS put into bins (below 0.5, 0.5-0.8, 0.8-1.0, above 1.0) plus the plan in service. The reward is minus the mean number of halted vehicles. After 12 short training episodes it averaged 40.6 s delay on three unseen seeds against SCATS's 36.4 s: about 4 s worse.

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \cdot \max Q(s', \cdot) - Q(s, a) \right]$$

α = learning rate, γ = how much the future counts, s' = next state

$$\text{target} = r + \gamma \cdot \max Q(s', \cdot)$$

what this action now seems to be worth

SIMPLE CALCULATION · One Q-learning update

Current Q(s, SP2) = −5. The agent picks SP2 and sees reward r = −4 (four halted cars). The best value in the next state is −3. Learning rate α = 0.1, discount γ = 0.9.

1. Target: −4 + 0.9 × (−3) = −4 − 2.7 = **−6.7**.
2. Surprise (error): −6.7 − (−5) = **−1.7**.
3. Step: 0.1 × (−1.7) = **−0.17**.
4. New Q(s, SP2) = −5 − 0.17 = **−5.17**.

SP2 now looks slightly worse in this state, so the agent will try it a little less. Thousands of such small updates are needed; 12 short episodes give only a few hundred per state.

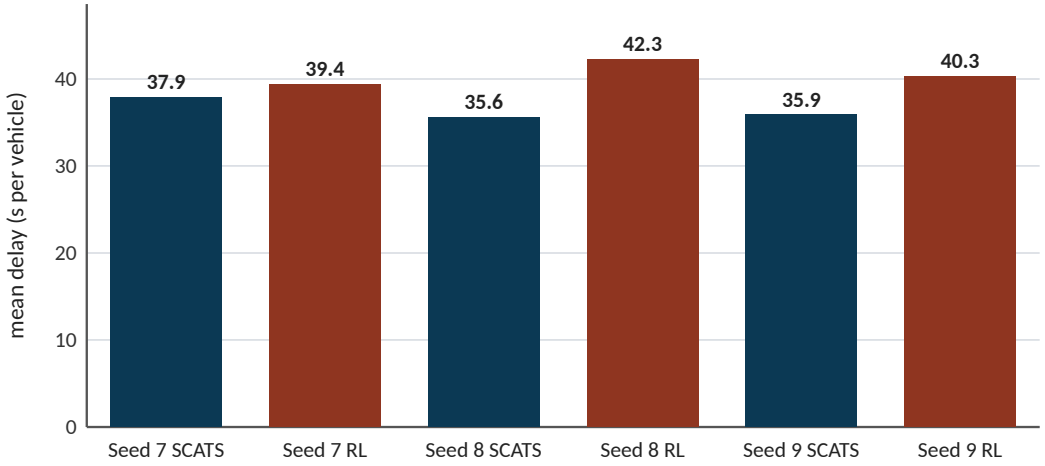


Figure 3. Mean delay on three unseen seeds: SCATS against the RL agent. What to notice: the agent (rust bars) is worse than SCATS (navy bars) on every seed, by 1.5 to 6.7 s. Means: SCATS 36.4 s, RL 40.6 s. From `rl_vs_scats.csv`.

TRAP

A losing result is not a useless result. The agent lost because its state is coarse, its reward is noisy, it trained for only 12 episodes, and SCATS's DS rule is a strong, well-tuned baseline. Each reason is a concrete next experiment.

PICTURE THIS · STOP 3 OF THE ROUTE

In the fish market a parrot sits in a cage holding exactly four menu cards. It may only shout a dish from those cards. When the queue at the counter shrinks it gets a sardine. It still shouts the wrong card so often that its queue is four people longer than the old fishmonger's.
What it encodes: Four cards = SP1 to SP4, the safe envelope; sardine = reward (fewer halted cars); four people longer = about 4 s worse (40.6 vs 36.4 s).

TRY IT NOW (about 90 seconds)

Q(s, a) = −2, reward −3, best next value −2, α = 0.2, γ = 0.9. What is the new Q(s, a)? *Answer in the key at the back.*

RECAP SO FAR

1. **Fair testing:** change one hook, compare with fixed, actuated and SCATS on the same seeds, report mean and spread of several measures, check deployability.
2. **DS from CVs:** $\hat{n} = n_{CV}/p$, $g' = \hat{n}\cdot 3600/MF$, DS = g'/g; noise $\propto 1/\sqrt{p}$; error 0.65 at 5%, 0.08 at 100%; closed-loop delay +8% to +21%.
3. **Learning splits: a Q-learning agent picks only approved plans SP1-SP4 (safe envelope); $Q \leftarrow Q + \alpha[r + \gamma \max Q' - Q]$; after 12 episodes RL 40.6 s vs SCATS 36.4 s.**

CONCEPT 4 OF 6 · SCATS MASTERY

SCATS and max-pressure

HOOK

SCATS's DS only looks at the stop line. If the road beyond the junction is already full, DS still asks for more green, for cars that have nowhere to go.



IN PLAIN WORDS

Think of doors between crowded rooms. Open the door where many people wait on your side and few on the other side. Opening a door into an already packed room just moves the crush. *Where the picture breaks:* people in rooms do not travel in platoons; cars do, and pure max-pressure ignores the timing of platoons, which is why it hurts green waves and buses.

Max-pressure control (Varaiya, 2013) scores each phase by its **pressure**: the queue waiting to use it minus the load on the road it feeds. It serves the phase with the highest pressure. Under ideal assumptions it is proven to carry as much traffic as any controller can.

Pure max-pressure is **acyclic**: it does not keep a fixed cycle, so neighbouring junctions cannot keep a **green wave** (offsets). The lab's **hybrid** keeps SCATS's cycle length and offsets and only blends the greens: a share β comes from pressure, the rest from SCATS's DS-based split.

On this lab's short, uncongested corridor SCATS still won: 36.9 s mean delay, against 40.8 s for pure max-pressure and 40.2 s for the hybrid with β = 0.5. Pure max-pressure stopped vehicles more (1.75 stops per vehicle against 1.30) and nearly doubled bus delay (57.4 s against 29.7 s), because it ignores progression.

$$\text{pressure}_p = \Sigma\ (\ \text{queue}_{\text{upstream}} - ½ \cdot \text{load}_{\text{downstream}}\ )$$

summed over the lanes the phase serves

$$g_p = (1 - \beta) \cdot g_{\text{SCATS},p} + \beta \cdot \text{avail} \cdot \text{pressure}_p \div \Sigma\ \text{pressure}$$

β = 0: pure SCATS; β = 1: greens from pressure only

SIMPLE CALCULATION · Blending 70 s of green between two phases

Phase A: 12 cars queued, 16 on its exit road. Phase B: 8 queued, 2 on its exit. 70 s of green to share. SCATS's own split gives A 40 s, B 30 s. β = 0.5.

1. Pressure A = 12 − ½ × 16 = **4**. Pressure B = 8 − ½ × 2 = **7**. Total 11.
2. Pressure greens: A = 70 × 4 ÷ 11 = **25.5 s**; B = 70 × 7 ÷ 11 = **44.5 s**.
3. Blend A: 0.5 × 40 + 0.5 × 25.5 = **32.7 s**.
4. Blend B: 0.5 × 30 + 0.5 × 44.5 = **37.3 s**. Check: 32.7 + 37.3 = 70 s.

Pressure pulls about 7 s away from A, whose exit is filling up. On a congested grid that prevents spill-back; on an uncongested corridor it just disturbs a good SCATS split.

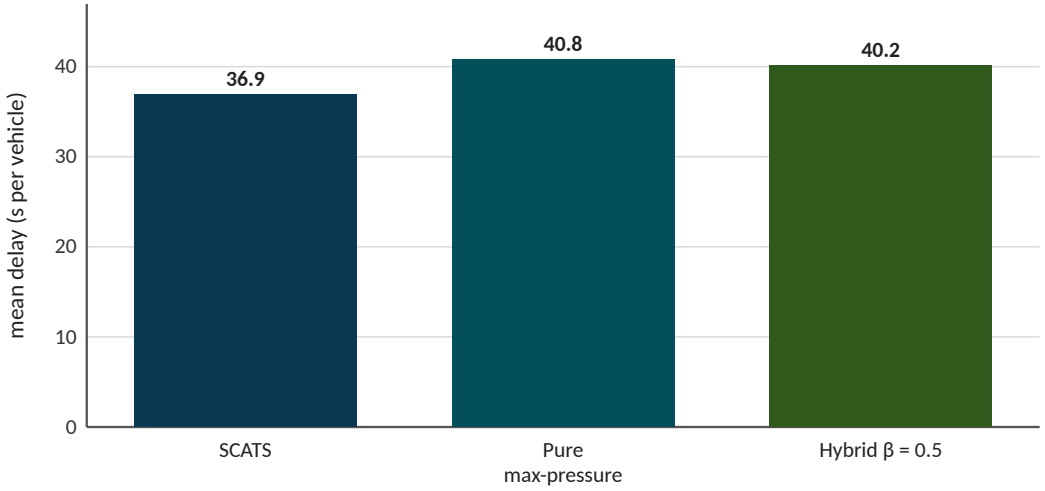


Figure 4. Mean delay on the corridor: SCATS, pure max-pressure and the hybrid. What to notice: both pressure-based methods lose on this short, uncongested corridor. Bus delay (not shown) was 29.7 s for SCATS, 57.4 s for pure max-pressure and 33.1 s for the hybrid. From `maxpressure.csv`.

TRAP

Max-pressure being 'throughput-optimal' does not mean 'lowest delay everywhere'. The proof assumes idealised queues and no platoons. Test it where it should shine: a grid with spill-back and turning conflicts.

PICTURE THIS · STOP 4 OF THE ROUTE

At the harbour locks two gates hold back boats. A pressure gauge on each gate shows boats waiting minus half the boats crammed in the basin beyond. The lock keeper turns a dial marked β halfway, mixing his old timetable with the gauges. Meanwhile the water bus, ignored, waits so long it sinks to its chimney.

What it encodes: Gauge = pressure (queue − ½ downstream load); half-turned dial = β = 0.5 blend; the sinking water bus = bus delay nearly doubling under pure max-pressure (57.4 vs 29.7 s).

TRY IT NOW (about 90 seconds)

60 s of green. Phase A: queue 10, downstream 4. Phase B: queue 6, downstream 12. SCATS split 30 s / 30 s, β = 0.5. Find the blended greens. *Answer in the key at the back.*

RECAP SO FAR

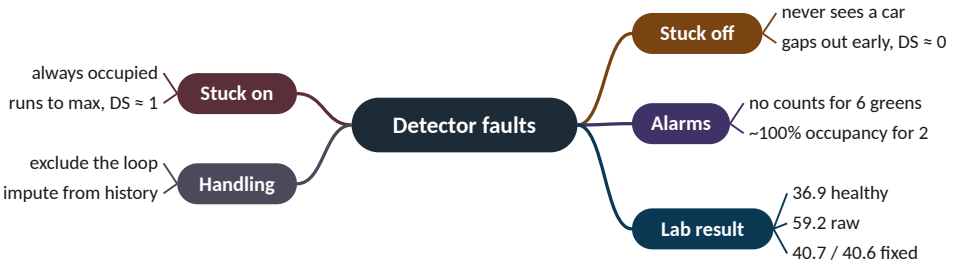
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3. **Learning splits:** a Q-learning agent picks only approved plans SP1-SP4 (safe envelope); $Q \leftarrow Q + \alpha[r + \gamma \max Q' - Q]$; after 12 episodes RL 40.6 s vs SCATS 36.4 s.
4. **Max-pressure: pressure = queue − ½ downstream load; hybrid keeps SCATS cycle and offsets, blends greens with β; corridor: SCATS 36.9, pure 40.8, hybrid 40.2 s.**

CONCEPT 5 OF 6 · SCATS MASTERY

Resilience to detector faults

HOOK

Every real city has broken loops. In the lab, when 3 loops in 10 failed, delay rose by 60%. Two simple 'this looks wrong' rules recovered most of it.



IN PLAIN WORDS

Imagine a smoke alarm stuck silent and another that never stops beeping. The silent one says all is calm; the beeping one says the house is always on fire. A sensible person notices 'this alarm has beeped for two days' and ignores it until it is repaired. *Where the picture breaks:* ignoring a fire alarm is dangerous; ignoring a broken loop is safe because SCATS still has other loops and history for that phase.

A **stuck-off** loop never sees a vehicle. The tactical controller sees endless gaps, ends the green early (gap-out at minimum green), and the loop's DS reads near zero. A **stuck-on** loop is permanently occupied. There are no gaps, so the phase runs to its maximum, and DS reads as fully saturated. Both mislead the **strategic** layer, which then sets wrong cycle lengths and splits for the whole subsystem.

The lab's experiment failed 30% of loops at minute 20, half stuck-off and half stuck-on. Using the raw data raised mean delay from 36.9 to 59.2 s per vehicle, a 60% increase.

Two plausibility **alarms** catch most faults: a loop with no counts for 6 greens in a row, or near-100% occupancy for 2 greens. An alarmed loop is then either **excluded** (the phase uses its other loops) or **imputed** (its DS is replaced by its own recent history). Both brought delay back to about 40.7 s, recovering about 83% of the loss.

$$\mathrm{DS} = (g - (T - n \cdot h)) \div g$$

g = green, T = unoccupied time, n = vehicles, h = standard space time

$$\text{recovered} = (\text{raw} - \text{fixed}) \div (\text{raw} - \text{healthy})$$

share of the fault damage that handling undoes

SIMPLE CALCULATION · What two broken loops report on a 30 s green

Green g = 30 s, h = 1.6 s. Healthy loop 1: 6 vehicles, unoccupied for 15.6 s. Phase DS = highest DS among its loops.

1. Healthy loop: g' = 30 − (15.6 − 6 × 1.6) = 30 − 6.0 = 24 s; DS = 24 ÷ 30 = **0.80**.
2. Stuck-off loop: n = 0, T = 30, so DS = (30 − 30) ÷ 30 = **0**. The highest is still 0.80, but if every loop of a phase is stuck off, DS = 0.
3. Stuck-on loop: T = 0, so DS = (30 − 0) ÷ 30 = **1.00**. The phase reads 1.00, wrongly saturated.
4. Exclude the alarmed stuck-on loop: phase DS = **0.80** again.
5. Lab recovery: (59.17 − 40.70) ÷ (59.17 − 36.94) = 18.47 ÷ 22.23 = **0.83**.

One stuck-on loop is enough to make a lightly used phase look full. Excluding it restores the right DS, and in the lab such handling recovered 83% of the lost performance.

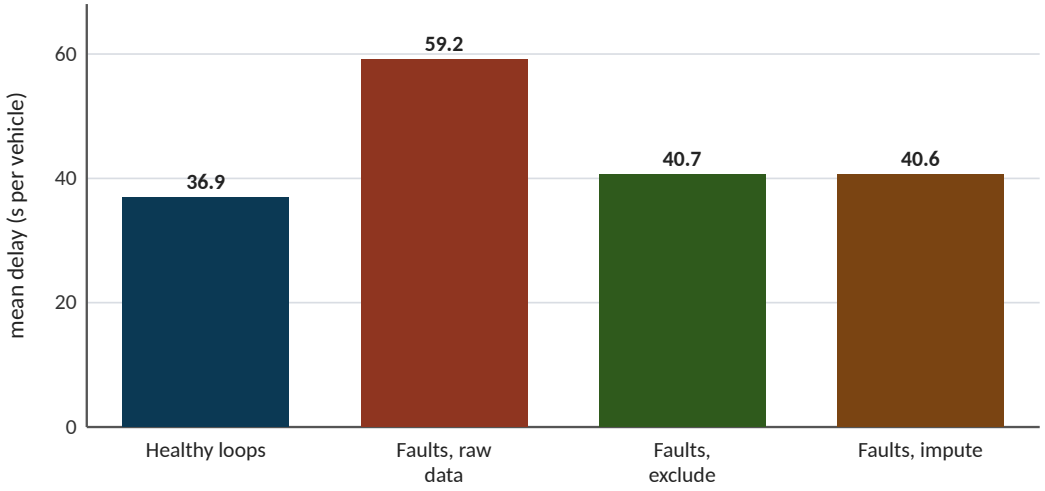


Figure 5. Mean delay with 30% of loops failed at minute 20. What to notice: raw faulty data raises mean delay from 36.9 to 59.2 s; excluding or imputing alarmed loops brings it back to 40.7 and 40.6 s. Stops barely change (1.30 to 1.40 per vehicle) and bus delay stays above healthy (29.7 vs 35-36 s). From `faults.csv`.

TRAP

Do not assume a faulty loop always shows up as 'zero'. A stuck-on loop is worse for the strategic layer: it looks like heavy traffic, so it can push the whole subsystem's cycle length up.

PICTURE THIS · STOP 5 OF THE ROUTE

In the bay floats a row of bell buoys. One bell is silent, wrapped in seaweed. Another rings non-stop, driving the gulls mad. A diver swims up and ties a red alarm flag to each, and the ringing one is muted. A delay balloon tied to the pier swells from 37 to 59, then shrinks back to 41.
What it encodes: Silent bell = stuck-off; ringing bell = stuck-on; red flags = alarms (6 greens of nothing, 2 greens of non-stop); balloon = delay 36.9 → 59.2 → 40.7 s.

TRY IT NOW (about 90 seconds)

A phase has two loops on a 40 s green. Loop 1 is healthy with DS 0.60; loop 2 is stuck on. What DS does SCATS use with raw data, and after excluding loop 2? *Answer in the key at the back.*

RECAP SO FAR

1. **Fair testing:** change one hook, compare with fixed, actuated and SCATS on the same seeds, report mean and spread of several measures, check deployability.
2. **DS from CVs:** $\hat{n} = n_{CV}/p$, $g' = \hat{n}\cdot 3600/MF$, DS = g'/g; noise $\propto 1/\sqrt{p}$; error 0.65 at 5%, 0.08 at 100%; closed-loop delay +8% to +21%.
3. **Learning splits:** a Q-learning agent picks only approved plans SP1-SP4 (safe envelope); $Q \leftarrow Q + \alpha[r + \gamma \max Q' - Q]$; after 12 episodes RL 40.6 s vs SCATS 36.4 s.
4. **Max-pressure:** pressure = queue − ½ downstream load; hybrid keeps SCATS cycle and offsets, blends greens with β; corridor: SCATS 36.9, pure 40.8, hybrid 40.2 s.
5. **Detector faults: stuck-off reads DS ≈ 0, stuck-on reads DS ≈ 1; alarms (6 empty greens, 2 full greens) then exclude or impute; delay 36.9 → 59.2 → 40.7 s.**

CONCEPT 6 OF 6 · SCATS MASTERY

Predicting signal timing for connected vehicles

HOOK

A car told 'the light turns red in 8 seconds' can ease off and save fuel. A fixed-time signal knows that exactly. An adaptive one changes its mind every cycle. How well can we predict it?



IN PLAIN WORDS

It is like a bus app predicting arrival. A timetable is easy to read but the driver may hurry or wait. The best guess is often the driver's own plan for today, not what the bus did yesterday. *Where the picture breaks:* a bus rarely skips a stop; SCATS can skip a phase or cut a green short when traffic stops, which no plan made at green start can know.

SPaT (Signal Phase and Timing) messages tell vehicles which light is showing and when it will change. They make **GLOSA** (green light optimal speed advice), eco-driving and automated driving possible. Fixed-time signals can publish exact times. SCATS re-plans every cycle and can **gap out** (end a green early when traffic stops) or skip phases, so the time to change is uncertain.

The lab logged every green on the corridor and tested three predictions made at the moment each green starts: the controller's **planned** green, **persistence** (the same green as last cycle) and a **linear regression** trained on the first half of the run.

They were compared by **mean absolute error (MAE)**: the average size of the miss, ignoring its sign. The plan won with 2.63 s; regression 3.24 s; persistence 3.94 s. The useful research lies in what a plan cannot know: live gap timers and priority calls.

$$\mathrm{MAE} = (|\mathrm{actual}_1 - \mathrm{predicted}_1| + \ldots + |\mathrm{actual}_n - \mathrm{predicted}_n|) \div n$$

average size of the error, in seconds

SIMPLE CALCULATION · Two predictors on four greens

Actual greens: 31, 26, 34, 29 s. Planned: 32, 30, 33, 30 s. Persistence (last cycle's green): 28, 31, 26, 34 s.

1. Planned errors: |31 − 32| = 1, |26 − 30| = 4, |34 − 33| = 1, |29 − 30| = 1.
2. Planned MAE = (1 + 4 + 1 + 1) ÷ 4 = 7 ÷ 4 = **1.75 s**.
3. Persistence errors: |31 − 28| = 3, |26 − 31| = 5, |34 − 26| = 8, |29 − 34| = 5.
4. Persistence MAE = (3 + 5 + 8 + 5) ÷ 4 = 21 ÷ 4 = **5.25 s**.

The plan wins because it already includes this cycle's decision. Persistence is always one cycle behind. The same ranking appeared in the lab (2.63 s against 3.94 s). Numbers here are illustrative.

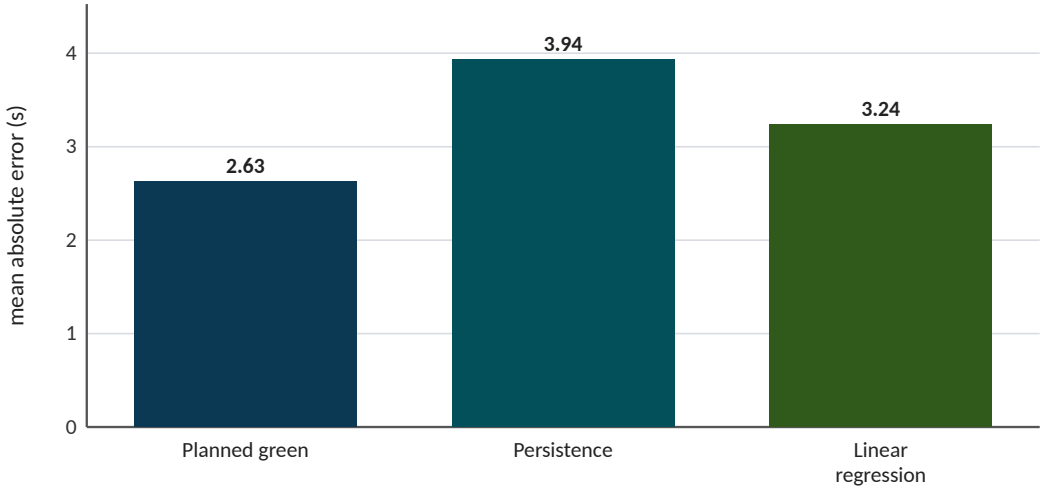


Figure 6. Green-duration prediction error on the SUMO corridor. What to notice: the controller's own plan is the best of the three; learning from history did not beat it. From `spat_mae.csv`.

TRAP

A low average error can hide large misses. A green that gaps out 10 s early can catch a car that trusted the prediction. Publish a confidence band, not just one number, and update it every second.

PICTURE THIS · STOP 6 OF THE ROUTE

At the ferry terminal a departures board shows the next sailing with a small error, about two and a half minutes. Beside it an old sailor shouts yesterday's times and is off by four. A third man with a calculator lands in between. A ferry suddenly leaves early because nobody is boarding.
What it encodes: Board = planned green (2.63 s); old sailor = persistence (3.94 s); calculator = regression (3.24 s); early ferry = gap-out, which no plan can foresee.

TRY IT NOW (about 90 seconds)

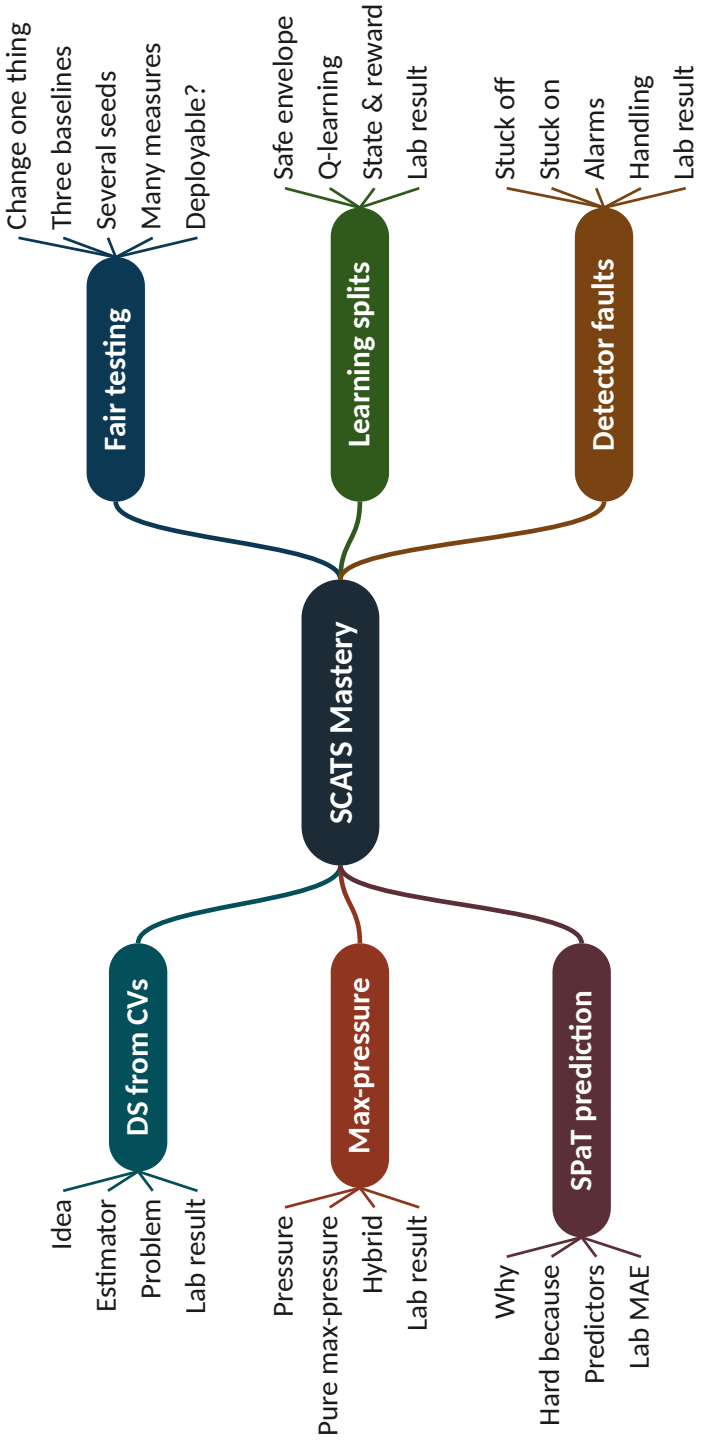
Two greens last 40 s and 35 s. The plan predicted 38 s for both. What is the MAE? *Answer in the key at the back.*

RECAP SO FAR

1. **Fair testing:** change one hook, compare with fixed, actuated and SCATS on the same seeds, report mean and spread of several measures, check deployability.
2. **DS from CVs:** $\hat{n} = n_{CV}/p$, $g' = \hat{n} \cdot 3600/MF$, $DS = g'/g$; noise $\propto 1/\sqrt{p}$; error 0.65 at 5%, 0.08 at 100%; closed-loop delay +8% to +21%.
3. **Learning splits:** a Q-learning agent picks only approved plans SP1-SP4 (safe envelope); $Q \leftarrow Q + \alpha[r + \gamma \max Q' - Q]$; after 12 episodes RL 40.6 s vs SCATS 36.4 s.
4. **Max-pressure:** pressure = queue − ½ downstream load; hybrid keeps SCATS cycle and offsets, blends greens with β; corridor: SCATS 36.9, pure 40.8, hybrid 40.2 s.
5. **Detector faults:** stuck-off reads DS ≈ 0, stuck-on reads DS ≈ 1; alarms (6 empty greens, 2 full greens) then exclude or impute; delay 36.9 → 59.2 → 40.7 s.
6. **SPaT prediction: SCATS re-plans and gaps out, so timing is uncertain; MAE compares predictors; planned 2.63 s beats regression 3.24 s and persistence 3.94 s.**

The master mind map

Every concept and its main branches on one page. Turn the page sideways. Try redrawing it from memory on blank paper, then compare.



Connections and contrasts

Five research directions at a glance

Direction	SCATS part replaced	Lab result	Beat SCATS?
CV-based DS	ds_source	delay +8% to +21%	no
RL split selection	split_selector	40.6 vs 36.4 s	no
Max-pressure hybrid	green_allocator	40.2 vs 36.9 s	no (corridor)
Fault alarms	sensor_filter / fault_handling	59.2 → 40.7 s	recovers 83% of fault loss
SPaT prediction	none (prediction only)	planned MAE 2.63 s	best of three predictors

Things beginners mix up

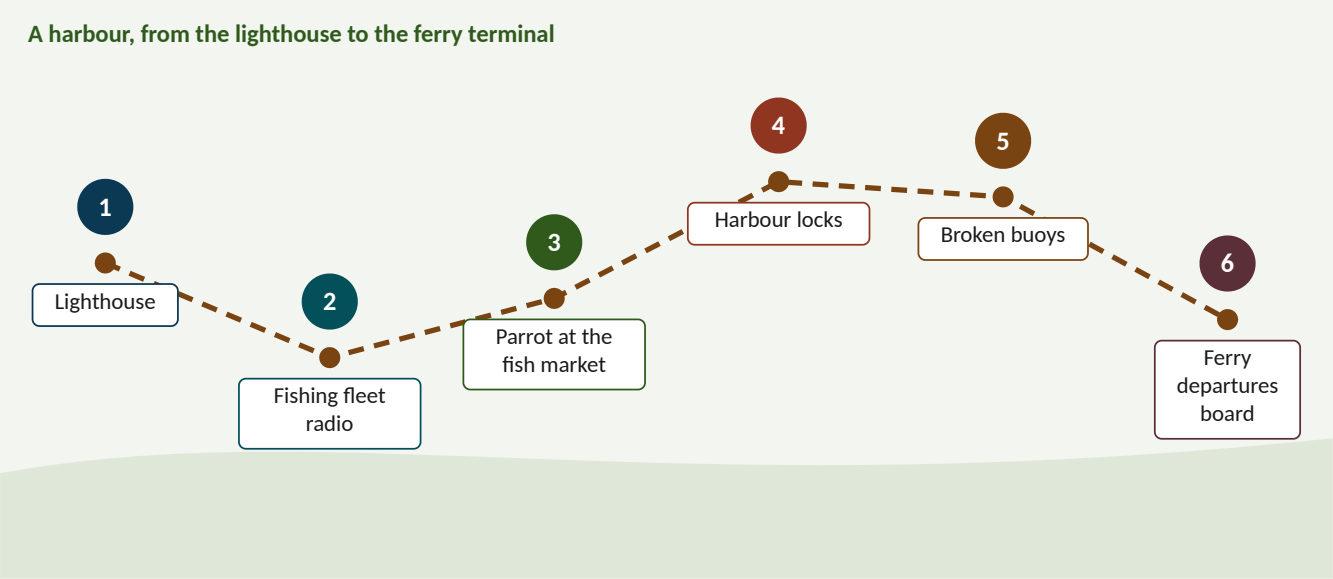
This...	...is not this	The one fact that tells them apart
Throughput-optimal	**Lowest delay**	Max-pressure's proof is about carrying the most traffic under ideal queues, not delay with platoons.
Stuck-off loop	**Stuck-on loop**	Stuck-off reads DS 0 and gaps out early; stuck-on reads DS 1 and runs to maximum.
Shadow mode	**Closed loop**	Shadow mode only compares estimates on the same traffic; closed loop lets the estimate drive the lights.
Penetration p	**Accuracy**	Error falls like 1/√p: four times the penetration halves the error.
Training seeds	**Test seeds**	Evaluate on seeds the method never trained on (here 7, 8, 9).

Symbols and terms

Symbol / term	Meaning
DS	degree of saturation = (g − (T − n·h)) ÷ g
p	connected-vehicle penetration rate (0 to 1)
MF	maximum flow of a lane, veh per hour of green (1,800 here)
$\hat{n}$, $\hat{g}$, $\hat{DS}$	estimated count, green needed and DS from CV data
Q(s, a)	learned value of action a in state s
α, γ	learning rate and discount factor in Q-learning
β	blend weight between SCATS greens and pressure greens
MAE	mean absolute error of a prediction
SPaT, GLOSA	signal phase and timing; green light optimal speed advice

The route: A harbour, from the lighthouse to the ferry terminal

Picture a busy harbour. You walk from the lighthouse on the point, along the quay, past the fish market and the buoys, to the ferry terminal. Each stop holds one idea. Make every scene loud, wet and silly.



The memory route. Walk it in order; each stop holds one concept.

Stop 1: Lighthouse (How to evaluate a new SCATS idea). At the lighthouse the keeper races four identical toy ships in a bathtub. She changes only one sail on the new ship, then throws three dice to make three different storms, and writes every result on the wall, even the races the new ship lost. *One sail = change one thing; four ships = three baselines plus the new idea; three dice = several seeds; writing every result = report mean, spread and losses.*

Stop 2: Fishing fleet radio (Degree of saturation from connected vehicles). On the quay, the harbourmaster counts returning fishing boats by radio. Only one boat in four has a radio, so every 'beep' he hears he multiplies by four on a huge chalkboard. A seagull steps on the radio, one extra beep sounds, and his total leaps by four boats. *Radio boats = connected vehicles; one in four = $p = 0.25$; multiply by four = $\hat{n} = n_{CV}/p$; the seagull's extra beep = sampling noise magnified by $1/p$.*

Stop 3: Parrot at the fish market (Learning split selection inside the SCATS envelope). In the fish market a parrot sits in a cage holding exactly four menu cards. It may only shout a dish from those cards. When the queue at the counter shrinks it gets a sardine. It still shouts the wrong card so often that its queue is four people longer than the old fishmonger's. *Four cards = SP1 to SP4, the safe envelope; sardine = reward (fewer halted cars); four people longer = about 4 s worse (40.6 vs 36.4 s).*

Stop 4: Harbour locks (SCATS and max-pressure). At the harbour locks two gates hold back boats. A pressure gauge on each gate shows boats waiting minus half the boats crammed in the basin beyond. The lock keeper turns a dial marked β halfway, mixing his old timetable with the gauges. Meanwhile the water bus, ignored, waits so long it sinks to its chimney. *Gauge = pressure (queue − ½ downstream load); half-turned dial = $\beta = 0.5$ blend; the sinking water bus = bus delay nearly doubling under pure max-pressure (57.4 vs 29.7 s).*

Stop 5: Broken buoys (Resilience to detector faults). In the bay floats a row of bell buoys. One bell is silent, wrapped in seaweed. Another rings non-stop, driving the gulls mad. A diver swims up and ties a red alarm flag to each, and the ringing one is muted. A delay balloon tied to the pier swells from 37 to 59, then shrinks back to 41. *Silent bell = stuck-off; ringing bell = stuck-on; red flags = alarms (6 greens of nothing, 2 greens of non-stop); balloon = delay 36.9 → 59.2 → 40.7 s.*

Stop 6: Ferry departures board (Predicting signal timing for connected vehicles). At the ferry terminal a departures board shows the next sailing with a small error, about two and a half minutes. Beside it an old sailor shouts yesterday's times and is off by four. A third man with a calculator lands in between. A ferry suddenly leaves early because nobody is boarding. *Board = planned green (2.63 s); old sailor = persistence (3.94 s); calculator = regression (3.24 s); early ferry = gap-out, which no plan can foresee.*

WALK-THROUGH DRILL

Close the guide. Walk the route in your head, saying each stop out loud and the one number or formula it holds. Then check against the cheat sheet. Repeat until you can do it in under two minutes.

Self-quiz

18 questions: warm-up (recall), core (use the ideas), challenge (combine them). Answer on paper first, then check the key.

Warm-up

Q1. Why should a new idea replace only one SCATS component in a test?

A. It runs faster
B. So any change in results is caused by that idea
C. SUMO only allows one change
D. To make SCATS look worse

Q2. What does a penetration rate p = 0.25 mean?

A. 25% of loops work
B. One vehicle in four is connected
C. 25 connected vehicles per hour
D. DS is 0.25

Q3. In this lab, what may the RL agent choose?

A. Any green length
B. The cycle length
C. One of four approved split plans
D. The offsets

Q4. A stuck-on loop makes a phase's DS look…

A. near zero
B. near one (saturated)
C. unchanged
D. negative

Q5. Which predictor of green duration had the lowest error in the lab?

A. Persistence
B. Linear regression
C. The controller's planned green
D. A coin toss

Core

Q6. p = 0.2, green 30 s, MF 1,800 veh/h, 3 CVs crossed. Estimated DS?

A. 0.20
B. 0.50
C. 1.00
D. 1.50

Q7. To halve the CV-based DS error, roughly how much must penetration grow?

A. ×1.4
B. ×2
C. ×4
D. ×10

Q8. Q = −4, r = −2, best next Q = −4, α = 0.5, γ = 0.5. New Q?

A. −3.0
B. −4.0
C. −5.0
D. −2.0

Q9. Mean delays on three unseen seeds were SCATS 36.4 s and RL 40.6 s. The best summary is:

A. RL is better
B. RL is about 4 s worse, consistently across seeds
C. No difference
D. RL is 40% worse

Q10. Phase A: queue 9, downstream 10. Phase B: queue 5, downstream 0. Which has the higher pressure?

A. A (4)
B. B (5)
C. Equal
D. Cannot tell

Q11. Why did pure max-pressure nearly double bus delay on the corridor?

A. It has no bus priority and no fixed cycle, so it breaks green waves
B. Buses are heavier
C. Pressure counts buses as zero
D. The corridor had no buses

Q12. In the hybrid, what does β = 0 give?

A. Pure max-pressure
B. Pure SCATS greens
C. Fixed time
D. Half of each

Q13. Which alarm rules does the lab use for faulty loops?

A. Any count above 30
B. No counts for 6 greens, or ~100% occupancy for 2 greens
C. DS above 0.9
D. A camera check

Q14. Healthy 36.9 s, raw faults 59.2 s, with alarms 40.7 s. What share of the fault damage was recovered?

A. about 40%
B. about 60%
C. about 83%
D. 100%

Challenge

Q15. At 100% penetration the CV DS error was still 0.08. Why is it not zero?

A. Rounding
B. Counting crossings cannot see how the green was used (gaps, slow starts)
C. Some cars are counted twice
D. MF is always wrong

Q16. A team reports their controller beats SCATS by 5 s, tested on the seeds used to train it, one measure only. Which two fixes matter most?

A. Use unseen test seeds and report stops and bus delay too
B. Run it faster and use more colours
C. Remove the SCATS baseline
D. Report only the best seed

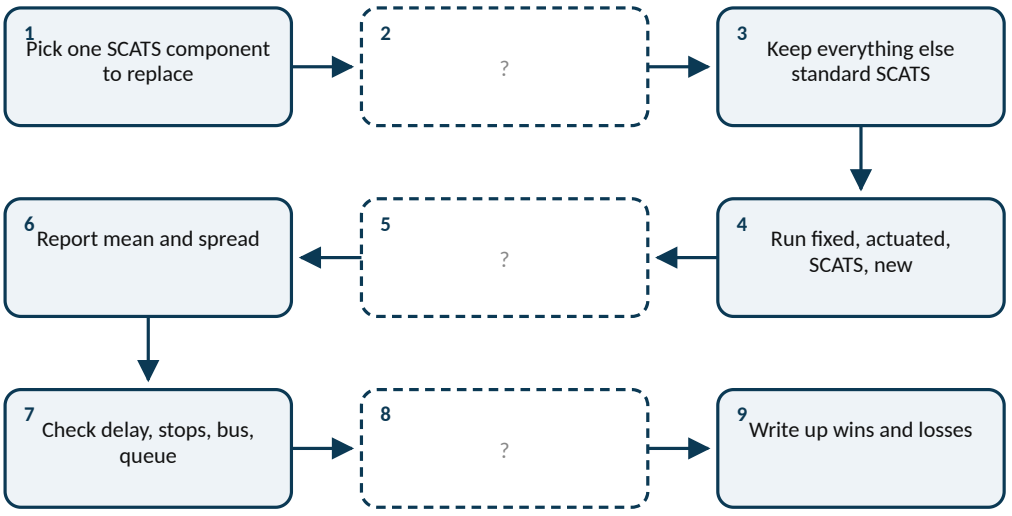
Q17. Where would max-pressure most likely beat SCATS?

A. A short, uncongested corridor
B. A congested grid with spill-back
C. An isolated junction at night
D. A road with no detectors

Q18. A GLOSA app uses the planned green (MAE 2.63 s). What is the main remaining risk?

A. Average error too large
B. Rare large misses from gap-out or priority calls
C. Cars cannot receive SPaT
D. Planned greens are random

Draw it. Fill the blank boxes of this flow from memory (it is Figure 1).



Where to go if you missed a question

Missed	Do this
Q1, Q16	Re-read concept 1: How to evaluate a new SCATS idea
Q2, Q6, Q7, Q15	Re-read concept 2: Degree of saturation from connected vehicles
Q3, Q8, Q9	Re-read concept 3: Learning split selection inside the SCATS envelope
Q10, Q11, Q12, Q17	Re-read concept 4: SCATS and max-pressure
Q4, Q13, Q14	Re-read concept 5: Resilience to detector faults
Q5, Q18	Re-read concept 6: Predicting signal timing for connected vehicles

Make it stick

BUILD THE SMALLEST WORKING EXAMPLE (15 min)

On paper, simulate CV sampling. Take 10 greens of 12 vehicles each (green 30 s, 2 s per vehicle, so true DS = 0.80). For each green, flip a coin for every vehicle twice: count it as connected only if both flips are heads (p = 0.25). Compute $D\hat{S}$ for each green and the mean absolute error.

Self-check: Each green's $D\hat{S} = n_{CV} \times 4 \times 2 \div 30 = 0.267 \times n_{CV}$. You expect $n_{CV} = 3$ on average ($D\hat{S}$ 0.80), but a spread of about ±1.5, so typical errors of 0.3 to 0.4, close to the lab's 0.34 at 25%.

TEACH IT TO A 14-YEAR-OLD (10 min)

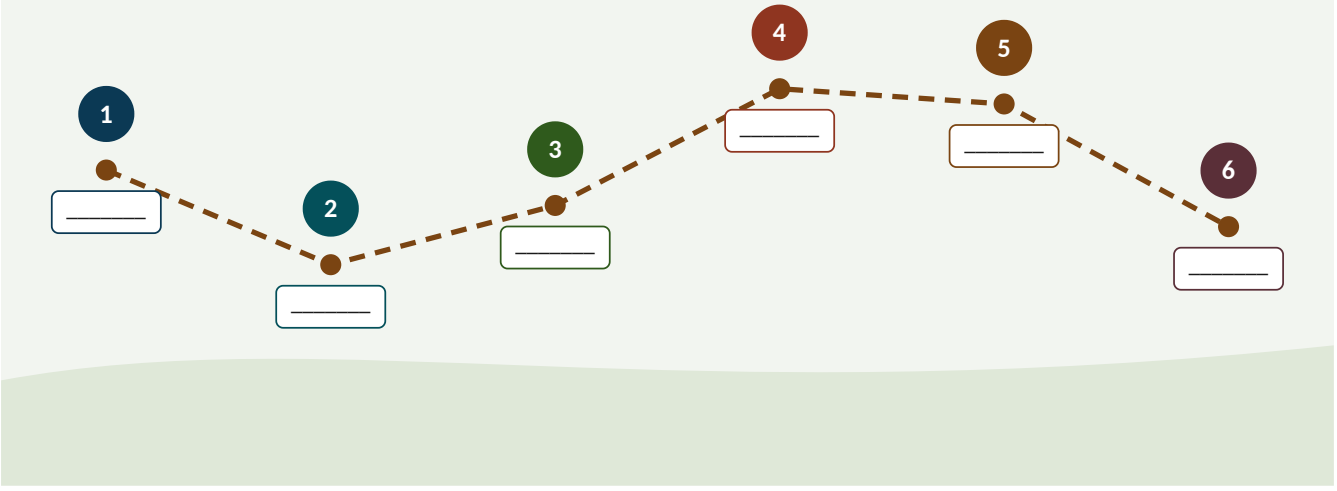
Explain to a 14-year-old why a new traffic-light idea that won one computer test might still be worse, and how you would test it properly.

Words you may not use: seed, baseline, penetration, saturation, reinforcement, algorithm

Fill the blank route

Write the concept held at each stop, and its key number.

A harbour, from the lighthouse to the ferry terminal



One-page cheat sheet

1. How to evaluate a new SCATS idea

One hook (ds_source, split_selector, green_allocator, sensor_filter). Baselines: fixed, actuated, SCATS. Several seeds; mean ± spread. Measures: delay, stops, bus delay, queue. RL vs SCATS: +1.46, +6.70, +4.41 → mean +4.19 s.

Route: Lighthouse

2. Degree of saturation from connected vehicles

$\hat{n} = n_{CV}/p$; $g^{c} = \hat{n}\cdot 3600/MF$; $D\hat{S} = g^{c}/g$. Error ∝ $1/\sqrt{p}$. Shadow MAE: 0.65 (5%), 0.34 (25%), 0.20 (50%), 0.08 (100%). Closed loop: delay 36.9 → 40.0-44.5 s.

Route: Fishing fleet radio

3. Learning split selection inside the SCATS envelope

Envelope: agent picks SP1-SP4 only; SCATS keeps cycle, offsets, min greens. State: DS bins (0.5, 0.8, 1.0) + plan. Reward: −halted vehicles. $Q \leftarrow Q + \alpha[r + \gamma\cdot maxQ' - Q]$. Q −5, r −4, maxQ' −3, α 0.1, γ 0.9 → −5.17. Result: RL 40.6 vs SCATS 36.4 s.

Route: Parrot at the fish market

4. SCATS and max-pressure

Pressure = $\Sigma(queue_{up} - ½\ load_{down})$. $g = (1-\beta)g_{SCATS} + \beta\cdot avail\cdot P/\Sigma P$. Example: A 12/16, B 8/2 → P 4 and 7 → 25.5/44.5 s → blend 32.7/37.3 s. Delay: SCATS 36.9, pure 40.8, hybrid 40.2 s; bus 29.7 / 57.4 / 33.1 s.

Route: Harbour locks

5. Resilience to detector faults

Stuck-off: no cars, early gap-out, DS 0. Stuck-on: no gaps, runs to max, DS 1. Alarms: 0 counts for 6 greens; ~100% occupancy for 2 greens. Handling: exclude or impute. 30% loops failed: healthy 36.9, raw 59.2 (+60%), exclude 40.7, impute 40.6 s; 83% recovered.

Route: Broken buoys

6. Predicting signal timing for connected vehicles

SPaT = phase and time-to-change, for GLOSA and automated cars. MAE = mean |actual − predicted|. Lab: planned 2.63, regression 3.24, persistence 3.94 s. Next: update each second with live gap timers; publish a confidence band.

Route: Ferry departures board

Numbers to remember

- RL vs SCATS on 3 seeds: 40.6 vs 36.4 s (+4.2 s)
- CV DS error: 0.65 (5%), 0.34 (25%), 0.08 (100%)
- CV closed loop: delay +8% to +21%
- Corridor: SCATS 36.9, pure MP 40.8, hybrid 40.2 s
- Bus delay: SCATS 29.7, pure MP 57.4 s
- Faults: 36.9 → 59.2 → 40.7 / 40.6 s
- SPaT MAE: planned 2.63, regression 3.24, persistence 3.94 s

Answer key

Try it now

Try it 1 (How to evaluate a new SCATS idea): Mean = (−2.0 + 1.0) ÷ 2 = **−0.5 s**. One seed is better and one worse, and the spread is far bigger than the mean, so you **cannot** claim it is better. Run more seeds.

Try it 2 (Degree of saturation from connected vehicles): $\hat{n}$ = 9 ÷ 0.5 = 18 vehicles; g^{c} = 18 × 3600 ÷ 1800 = **36 s**; $D\hat{S}$ = 36 ÷ 30 = **1.20**.

Try it 3 (Learning split selection inside the SCATS envelope): Target = −3 + 0.9 × (−2) = **−4.8**. Error = −4.8 − (−2) = −2.8. New Q = −2 + 0.2 × (−2.8) = **−2.56**.

Try it 4 (SCATS and max-pressure): Pressure A = 10 − 2 = 8; B = 6 − 6 = 0. Pressure greens: A 60 s, B 0 s. Blend: A = 0.5 × 30 + 0.5 × 60 = **45 s**; B = 0.5 × 30 + 0.5 × 0 = **15 s**. (A real controller would still enforce B's minimum green.)

Try it 5 (Resilience to detector faults): Stuck on: T = 0, DS = (40 − 0) ÷ 40 = 1.00. Raw phase DS = max(0.60, 1.00) = **1.00**. After excluding loop 2: **0.60**.

Try it 6 (Predicting signal timing for connected vehicles): Errors |40 − 38| = 2 and |35 − 38| = 3. MAE = (2 + 3) ÷ 2 = **2.5 s**.

Self-quiz

Q1: B. Changing one hook isolates the cause. Speed and SUMO limits are not the reason. (Concept 1)

Q2: B. p is the share of vehicles that report their position. (Concept 2)

Q3: C. The envelope limits it to SP1-SP4; SCATS keeps cycle, offsets and minimum greens. (Concept 3)

Q4: B. Permanently occupied means no unoccupied time T, so DS = g/g = 1. (Concept 5)

Q5: C. Planned 2.63 s, regression 3.24 s, persistence 3.94 s. (Concept 6)

Q6: C. $\hat{n}$ = 3 ÷ 0.2 = 15; g' = 15 × 2 = 30 s; DS = 30 ÷ 30 = 1.00. (Concept 2)

Q7: C. Error ∝ 1/√p, so halving it needs √p doubled, i.e. p × 4. (Concept 2)

Q8: B. Target = −2 + 0.5 × (−4) = −4; error 0; Q stays −4.0. (Concept 3)

Q9: B. RL lost on all three seeds by 1.5-6.7 s, mean 4.2 s (about 11%, not 40%). (Concept 3)

Q10: B. A = 9 − 5 = 4; B = 5 − 0 = 5. B wins despite its shorter queue because its exit is empty. (Concept 4)

Q11: A. Acyclic control ignores progression and priority, so buses stop more (57.4 vs 29.7 s). (Concept 4)

Q12: B. $g = (1 - \beta)g_{SCATS} + \beta \cdot$(pressure share); β = 0 leaves SCATS's split. (Concept 4)

Q13: B. These two plausibility rules catch stuck-off and stuck-on loops. (Concept 5)

Q14: C. (59.2 − 40.7) ÷ (59.2 − 36.9) = 18.5 ÷ 22.3 ≈ 0.83. (Concept 5)

Q15: B. Loop DS uses unoccupied time; a count assumes saturated flow, so it misses wasted green. (Concept 2)

Q16: A. Training-seed testing flatters the method; one measure can hide trade-offs. (Concept 1)

Q17: B. Pressure's advantage is reacting to full downstream roads; that matters only when spill-back occurs. (Concept 4)

Q18: B. MAE hides tail errors; a green ending 10 s early can catch a driver who trusted the advice. (Concept 6)

Draw it: the completed flow is Figure 1.

Your revision plan

When	What	Done
Today	Read one concept a sitting. Do every Try-it-now on paper.	☐
Day 1	Walk the route from memory. Redraw the master mind map.	☐
Day 3	Warm-up and core quiz without notes. Re-read what you missed.	☐
Day 7	Whole quiz. Do the 'build the smallest example' task.	☐
Day 14	Teach-it task out loud. Cheat sheet from memory, then compare.	☐
Before an exam or project	Cheat sheet plus route walk only.	☐

Going further

- Sims, A.G. and Dobinson, K.W. (1980). The Sydney coordinated adaptive traffic (SCAT) system: philosophy and benefits. *IEEE Transactions on Vehicular Technology*, 29(2).
- Lowrie, P.R. (1982). The Sydney coordinated adaptive traffic system: principles, methodology, algorithms. IEE International Conference on Road Traffic Signalling, London.
- Varaiya, P. (2013). Max pressure control of a network of signalized intersections. *Transportation Research Part C*, 36.
- Wei, H., Zheng, G., Gayah, V. and Li, Z. (2019). A survey on traffic signal control methods. arXiv:1904.08117.
- Lopez, P.A. et al. (2018). Microscopic traffic simulation using SUMO. IEEE ITSC.

- This lab's website: Mastery page and scripts in `sumo/mastery/`, smrahmanbd.github.io/scats-learning-lab.

HONESTY NOTE

SCATS is owned by Transport for NSW. This guide describes an independent, educational re-implementation of published SCATS concepts; parameter values are illustrative, not official SCATS settings. Worked examples use small invented numbers chosen to be easy by hand. Lab results come from this repository's SUMO runs on a four-intersection corridor (seed 1, or seeds 7-9 for the RL test) and show the method, not what any real SCATS installation achieves. The CV error values at 25% and 50% are read from the lab's figure; the 5% and 100% values are exact.